Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 255 – Strength of Materials

Lecture 4: Bending In Beams I

By: Samuel Boahen (PhD)

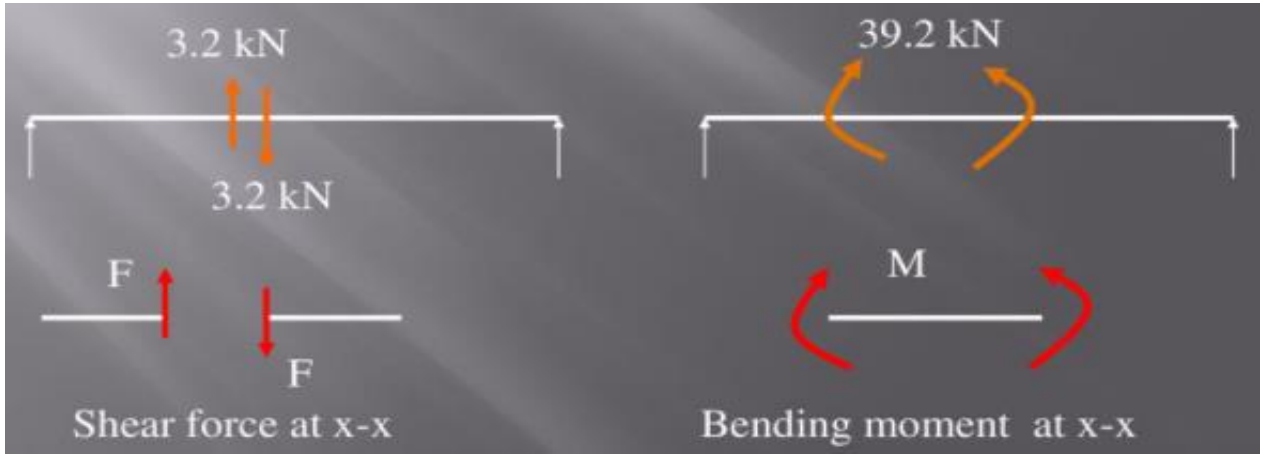
Lecture Outline

- Understand and calculate Shear Forces
- Understand and calculate Bending Moments
- Draw Shear Force Diagrams
- Draw Bending Moment Diagrams

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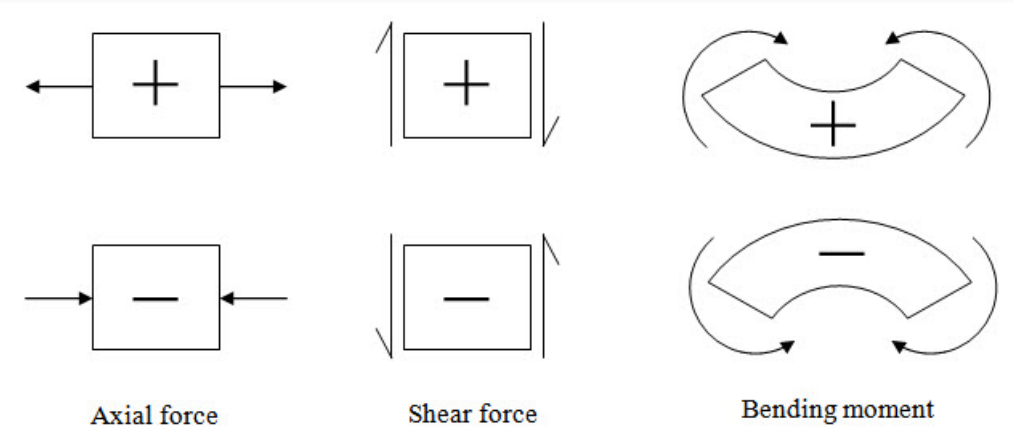
Shear Force and Bending Moment

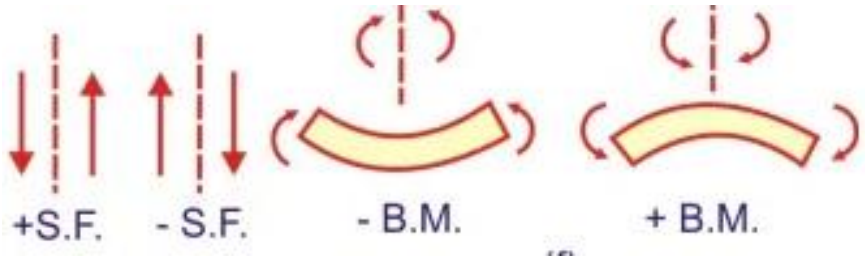
- **Shear force** at a section in a beam is the ***algebraic sum of vertical forces*** acting to the left or right of that section.
- **Bending moment** at a section is the ***algebraic sum of the moments of all forces*** acting to the left or right of the section.
- **Moment of a force** is equal to the ***force multiplied by the distance from a point.***



Shear and Bending In Beams

- Shear and bending moment diagrams are plots of the shear forces and bending moments respectively along the length of a beam.
- Sign conversions (positive or negative) for the shear force and bending moment in beams can be taken either ways, but must be maintained through a problem.





Shear Force and Bending Moment Diagrams

- A shear force diagram is one which shows the variation of the shear force along the length of the beam.
- A bending moment diagram is one which shows the variation of the bending moment along the length of the beam.
- Before drawing the shear force and bending moment diagrams, we must know the **different types of beams** and **different types of load acting** on the beams.

Types of Beams

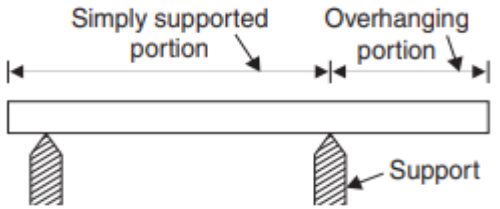
- **Cantilever beam**: A beam which is fixed at one end and free at the other end, is known as cantilever beam.
- **Simply supported beam**: A beam supported or resting freely on the supports at its both ends, is known as simply supported beam.
- **Overhanging beam**: If the end portion of a beam is extended beyond the support, such beam is known as overhanging beam.
- **Fixed beams:** A beam whose both ends are fixed or built-in walls, is known as fixed beam. A fixed beam is also known as a built-in or encastred beam.
- **Continuous beam**: A beam which has more than two supports is known as continuous beam.



Cantilever



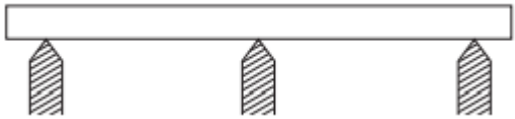
Simply supported beam



Overhanging beam



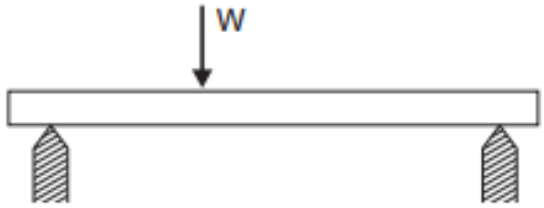
Fixed beam



Continuous beam

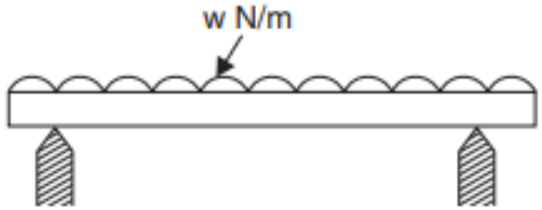
Types of Load

- **Concentrated or Point Load**: It is considered to act at a point, although in practice it must really be distributed over a small area.



Point Load

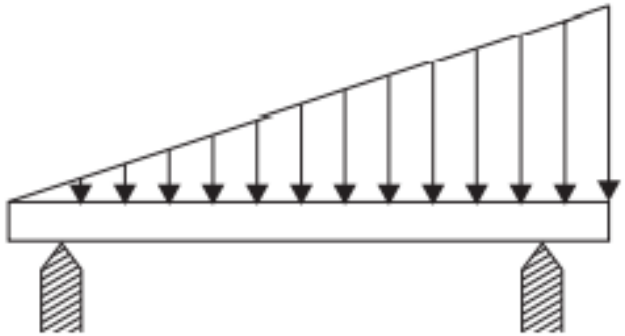
- **Uniformly Distributed Load**: A uniformly distributed load is one which is spread over a beam in such a manner that the rate of loading W is uniform along the length. For solving numerical problems, the total uniformly distributed load is converted into a point load, acting at the centre of uniformly distributed load.



Uniformly Distributed Load (u.d.l)

Types of Load

- **Uniformly Varying Load**: A uniformly varying load is one which is spread over a beam in such a manner that the rate of loading varies from point to point along the beam in which load is zero at one end and increases uniformly to the other end. Such load is known as triangular load.



Uniformly Varying Load

Shearing Force and Bending Moment Diagram

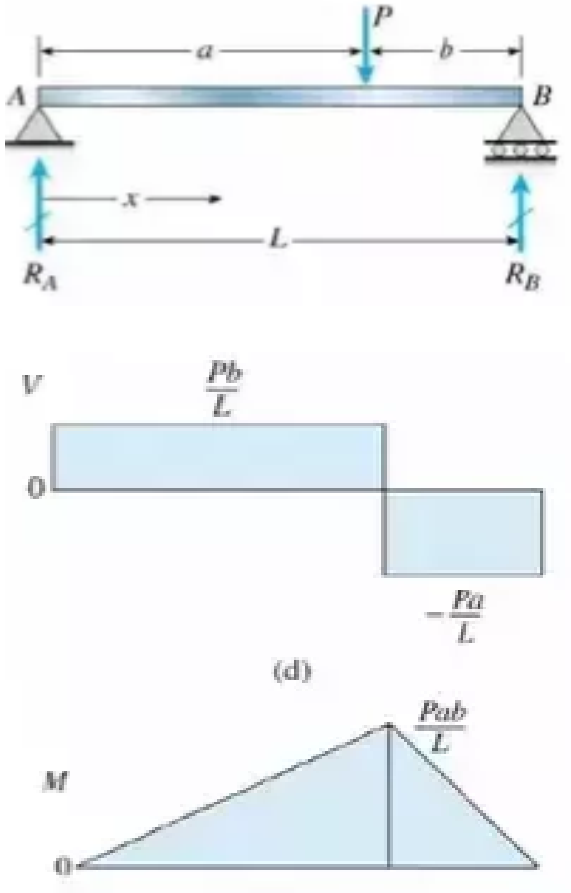
- Firstly, draw the free body diagram and determine the reactions using moment equations and sum of forces.
- Calculate the bending moment along the beam with emphasis on the extreme and critical points.
- Cut the beam at an arbitrary point.
- Show the unknown shear force and bending moment on the cut using the positive sign convention
- Sum forces in the vertical direction to determine the unknown shear.
- Sum moments about the cut to determine the unknown moment.
- Use the calculated forces and moments in drawing the bending moment and shear force diagrams.

Shearing Force and Bending Moment Diagram

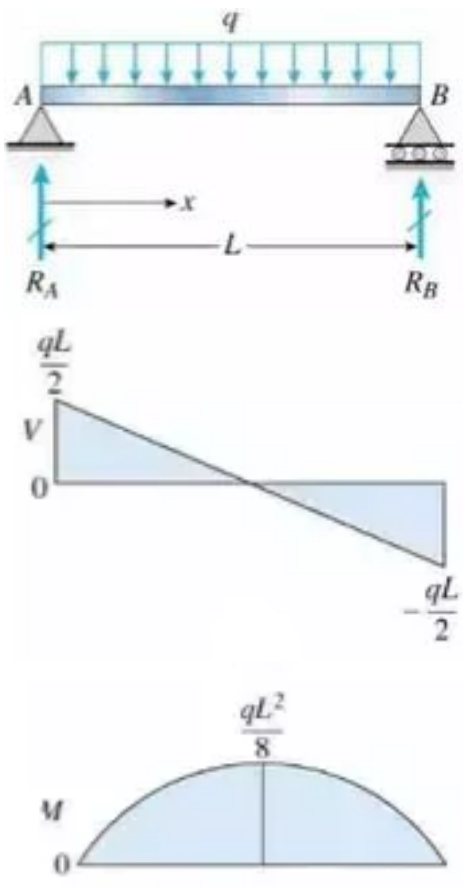
- Consider the left or the right portion of the section.
- Add the forces (including reaction) normal to the beam on one of the portion.
- The positive values of shear force and bending moments are plotted above the base line, and negative values below the base line.
- The shear force diagram will increase or decrease suddenly i.e., by a vertical straight line at a section where there is a vertical point load.
- The shear force between any two vertical loads will be constant and hence the shear force diagram between two vertical loads will be horizontal
- The bending moment at the two supports of a simply supported beam and at the free end of a cantilever will be zero.

Loads and their respective diagrams

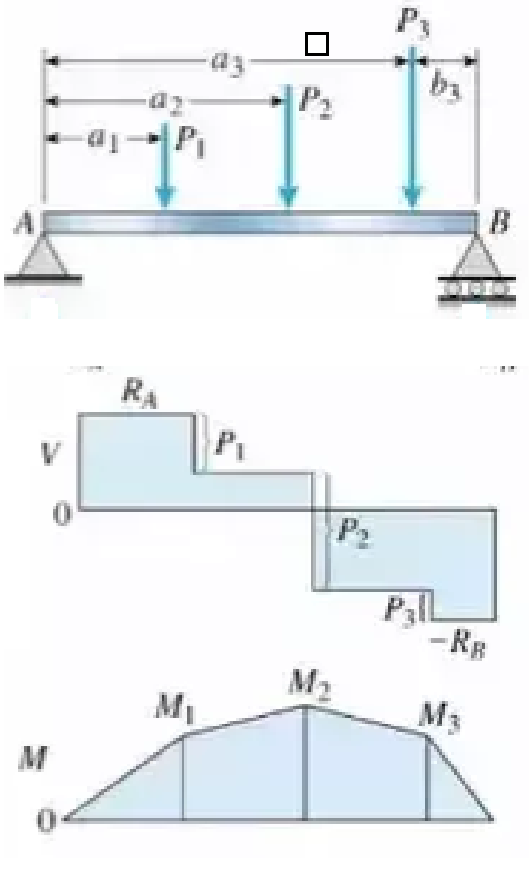
(a) Concentrated Load



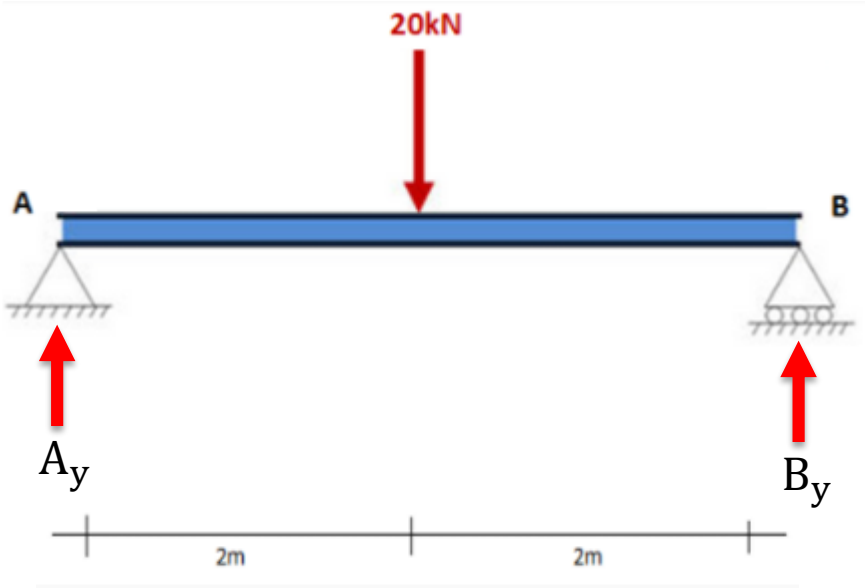
(b) Uniform Load



(c) Several Concentrated Load



Illustrative Example – Calculating Reactions at support



Steps

a) Let the sum of moments about a reaction point equal to ZERO ($\Sigma M = 0$).

$$\sum M_B = 0$$

$$20(2) - A_y(4) = 0$$

$$A_y = 10\ kN$$

b) Let the sum of vertical forces equal to 0 ($\Sigma F_y = 0$).

$$\Sigma F_Y = 0 = A_Y + B_Y - 20kN$$

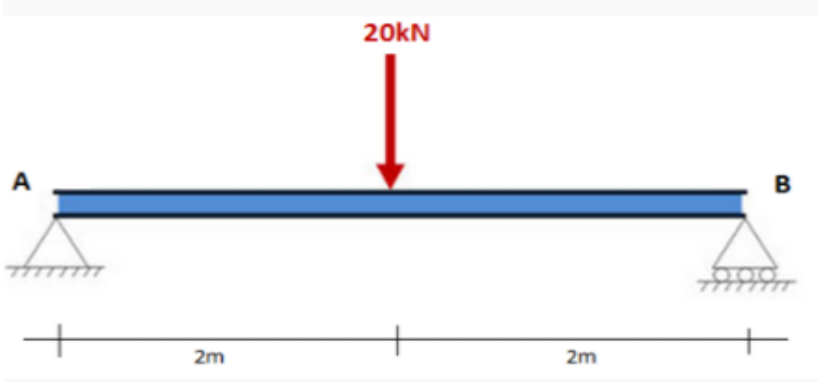
And since we know $A_Y = 10kN$

$$0 = 10kN + B_Y - 20kN$$

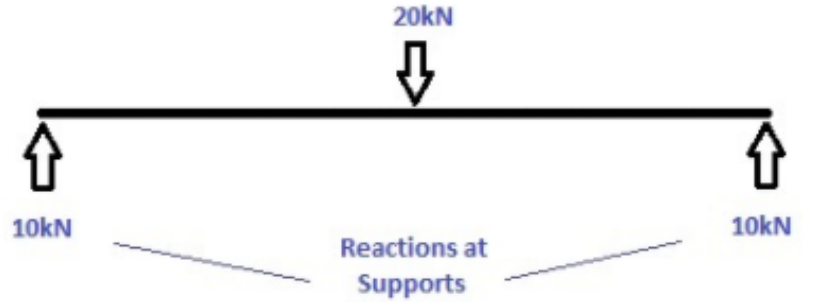
$$B_Y = 10kN$$

N:B: The sign convention we have chosen is that anti-clockwise moment are positive and clockwise moments are negative

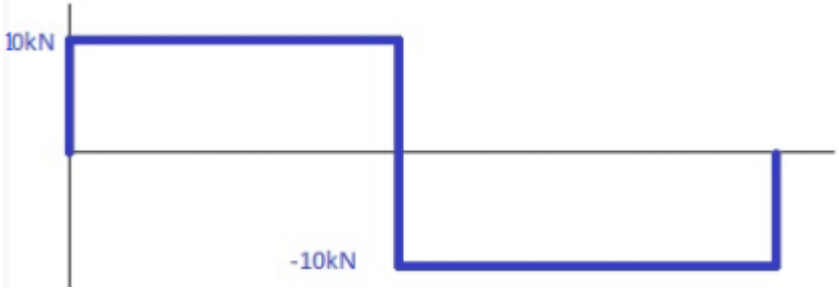
Illustrative Example – Drawing Free Body Diagram and Shear Force Diagram



b) Free body diagram



c) Shear Force diagram drawn using all vertical forces



Calculating Bending Moment

- To calculate the bending moment of a beam, we must work in the same way we did for the Shear Force Diagram.
- Starting at x = 0 we will move across the beam and calculate the bending moment at each point.

Moment at x=0

M(x) = -10(0)

M(x) = 0

Moment at x=2

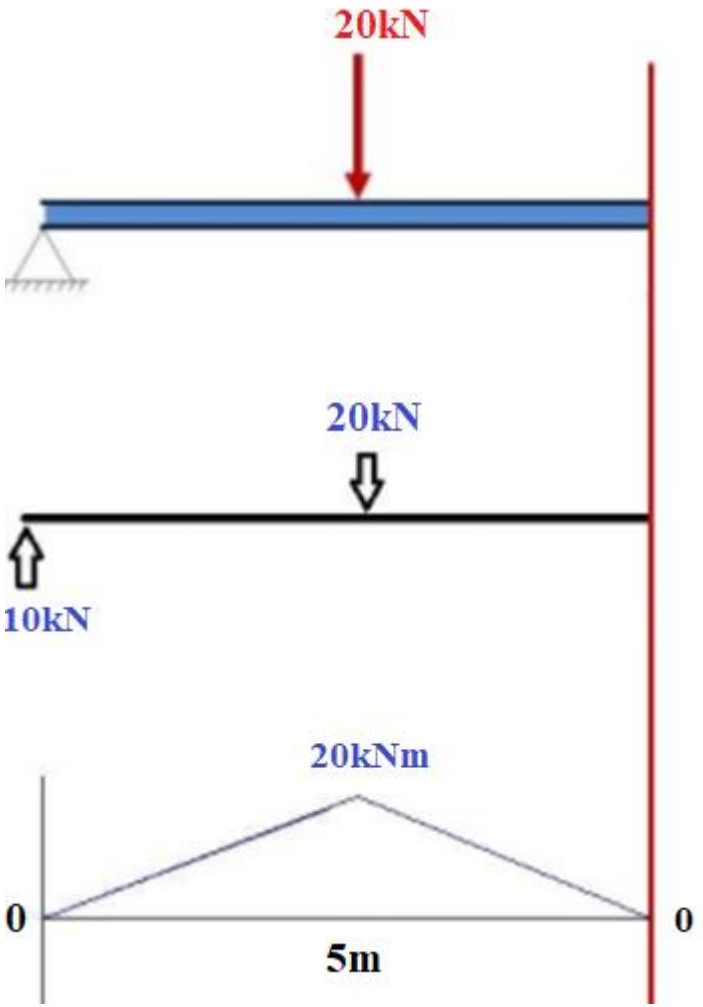
M(2) = 10(2)

M(2) = 20 kNm

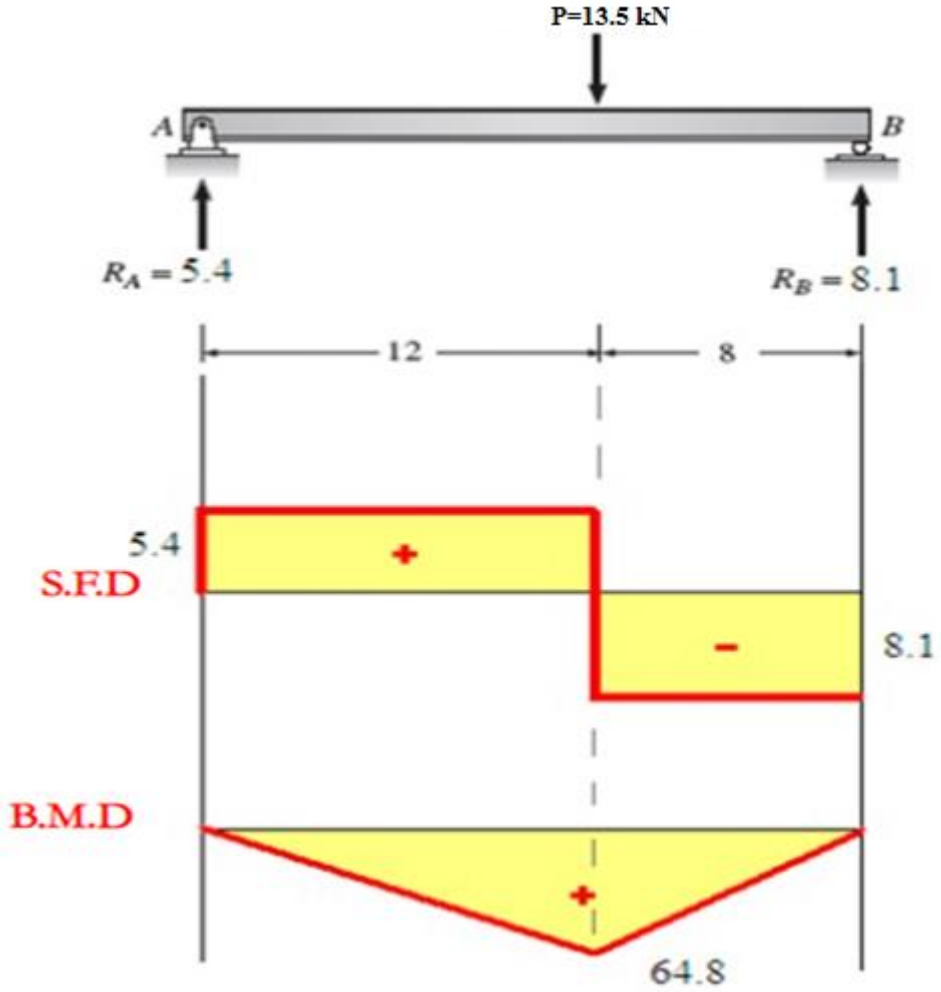
Moment at x=4

M(4) = 10(-4)+20(2)

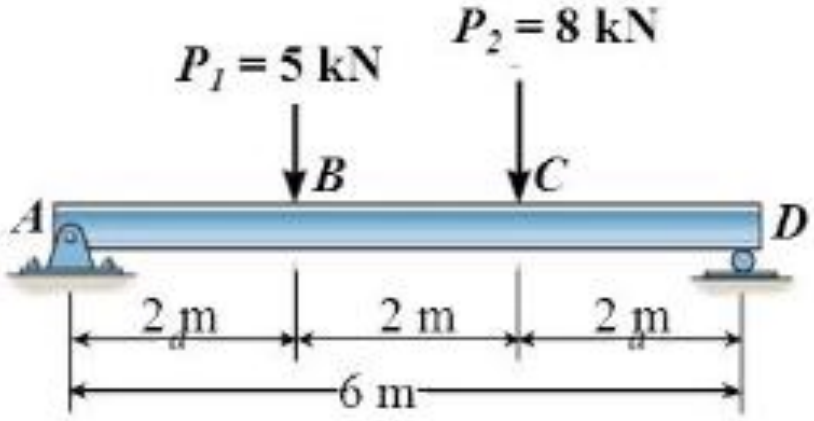
M(4) = 0 kNm



Shear force and Bending Moment Diagram

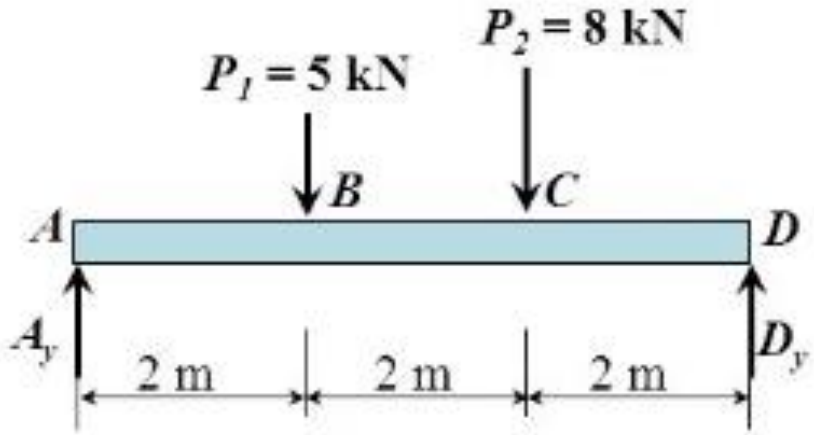


Example 1



$$\Sigma M_A = 0; \qquad D_y(6) - P_2(4) - P_1(2) = 0$$

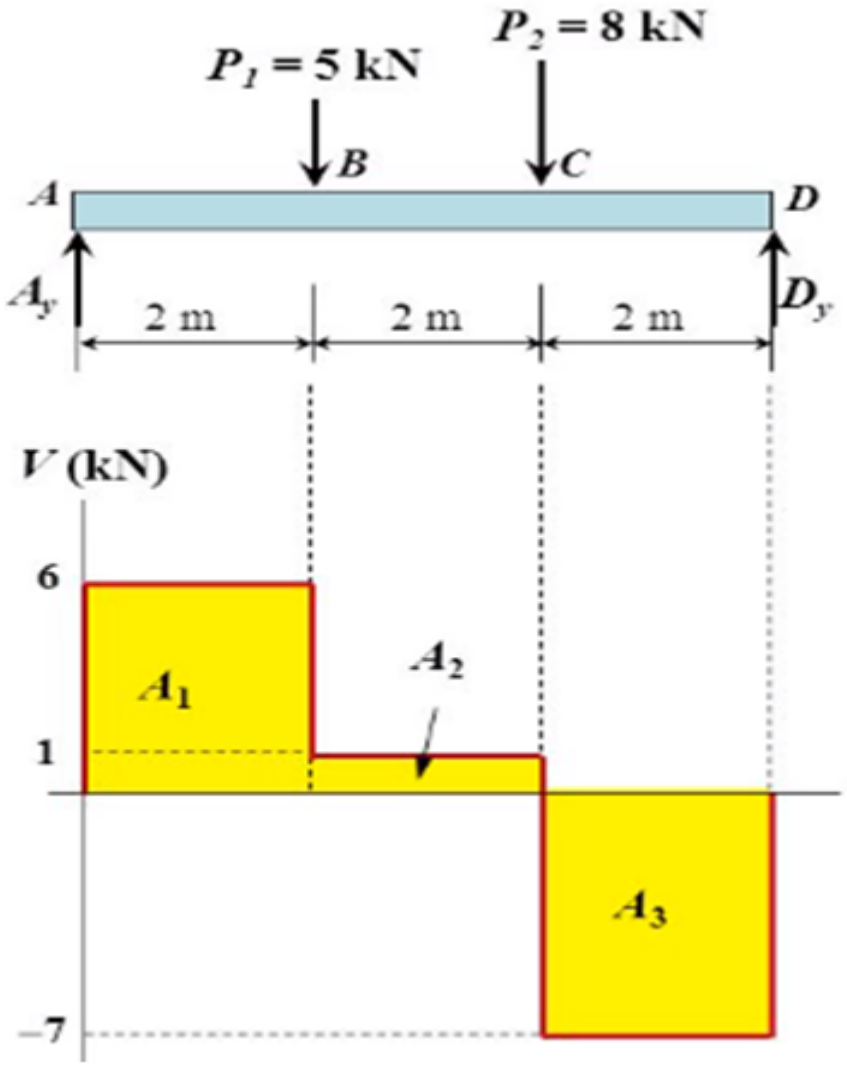
$$D_y = (32 + 10)/6 = 7 \text{ kN}$$



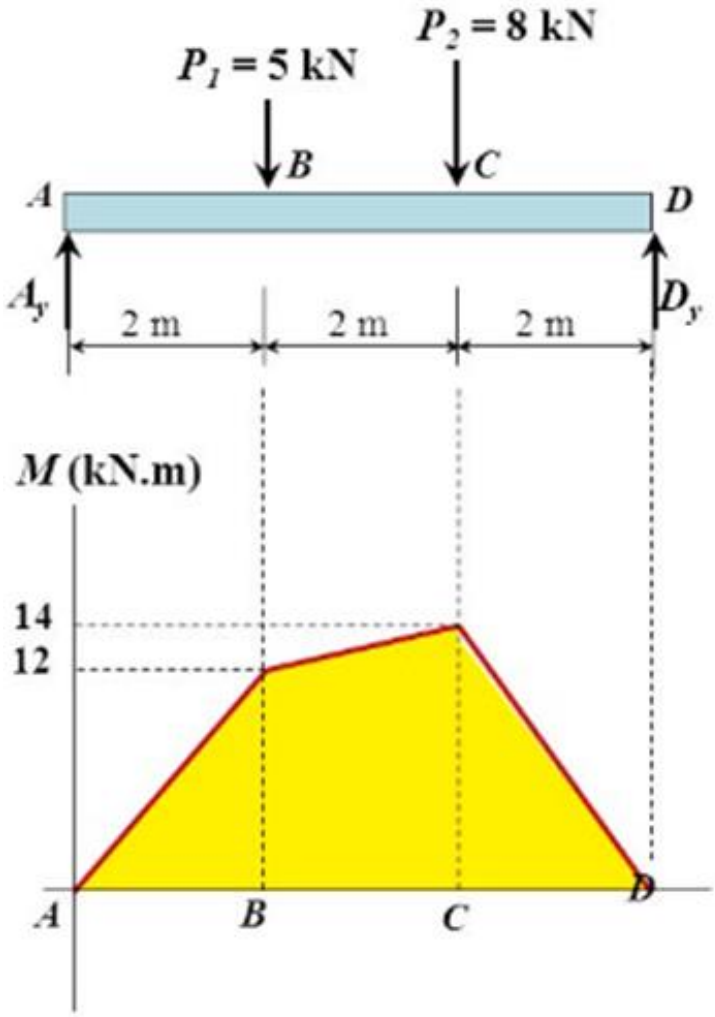
$$\Sigma F_y = 0; \qquad A_y - P_1 - P_2 + D_y = 0$$

$$A_y = P_1 + P_2 - D_y = 6 \text{ kN}$$

Example 1 (Shear Force Diagram)



Example 1



BENDING MOMENT DIAGRAM

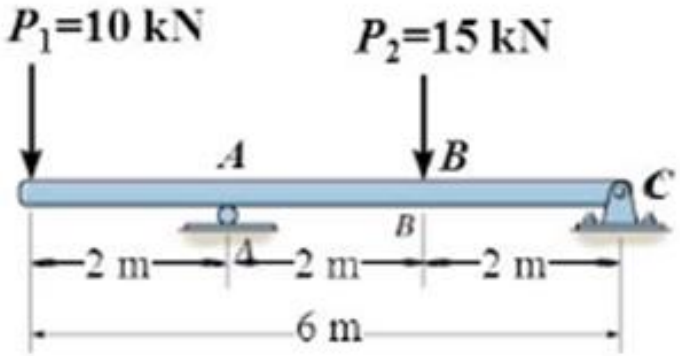
$M_0 = 0$

$M_2 = M_0 + A_1 = 0 + 12 = 12$ kN.m

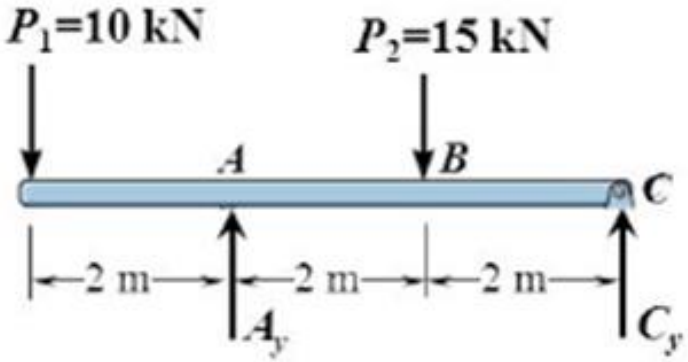
$M_4 = M_2 + A_2 = 12 + 2 = 14$ kN.m

$M_6 = M_4 + A_3 = 14 - 14 = 0$

Example 2



FREE BODY DIAGRAM and SUPPORT REACTIONS



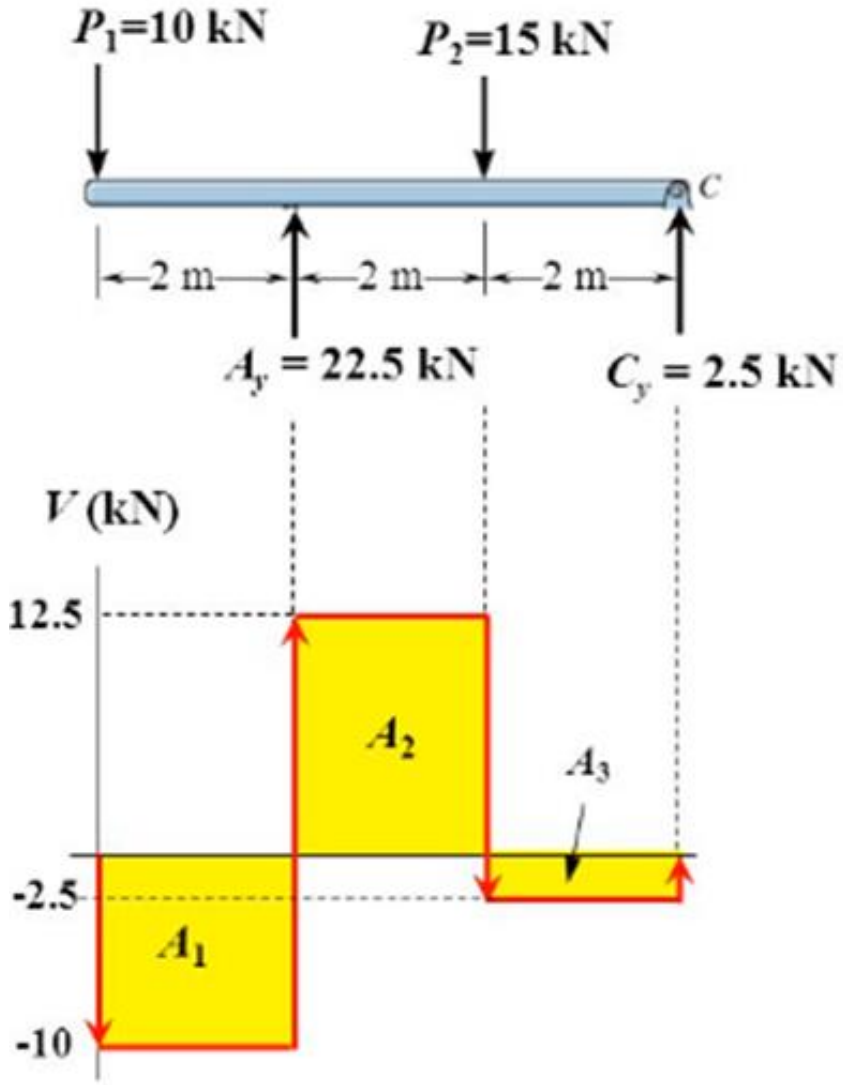
$$\Sigma M_A = 0; \qquad P_1(2) - P_2(2) + C_y(4) = 0$$

$$C_y = (30 - 20)/4 = 2.5 \text{ kN}$$

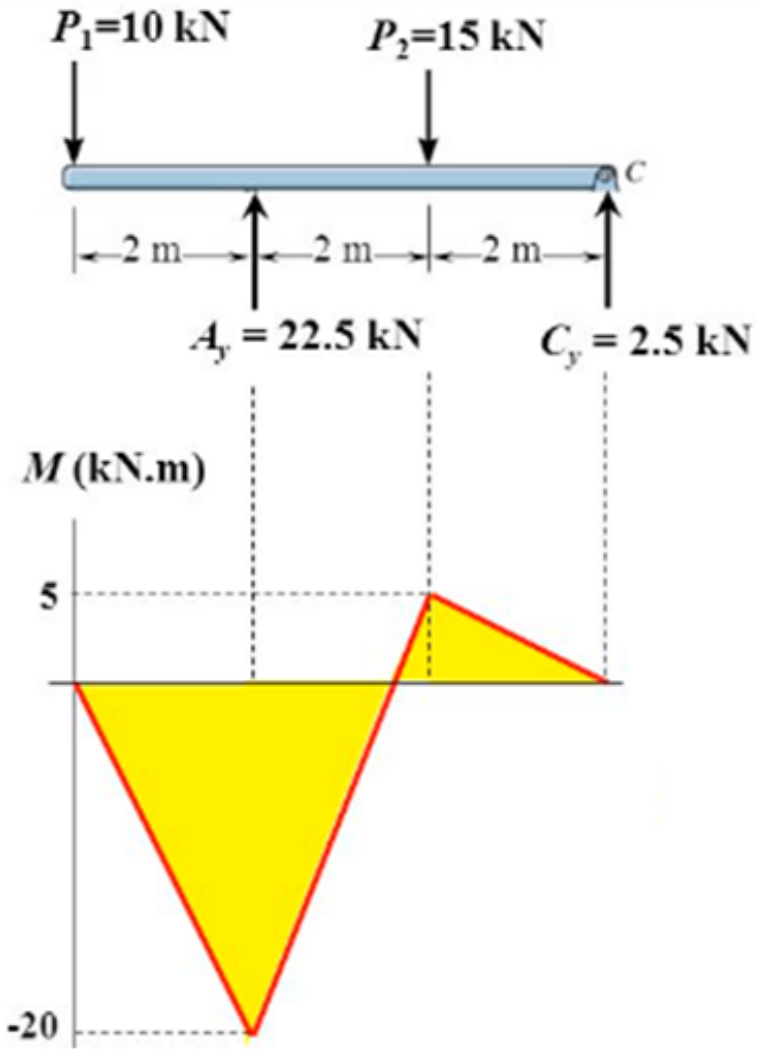
$$\Sigma F_y = 0; \qquad A_y - P_1 - P_2 + C_y = 0$$

$$A_y = 22.5 \text{ kN}$$

Example 2 (Shear Force Diagram)



Example 2



BENDING MOMENT DIAGRAM

$$M_0 = 0$$

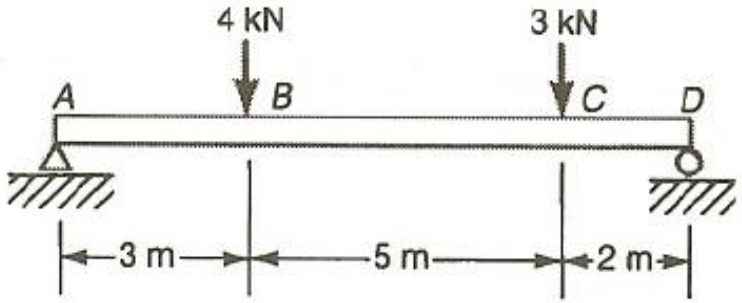
$$M_2 = M_0 + A_1$$
$$= 0 - 20 = -20 \text{ kN.m}$$

$$M_4 = M_2 + A_2$$
$$= -20 + 25 = 5 \text{ kN.m}$$

$$M_6 = M_4 + A_3$$
$$= 5 - 5 = 0$$

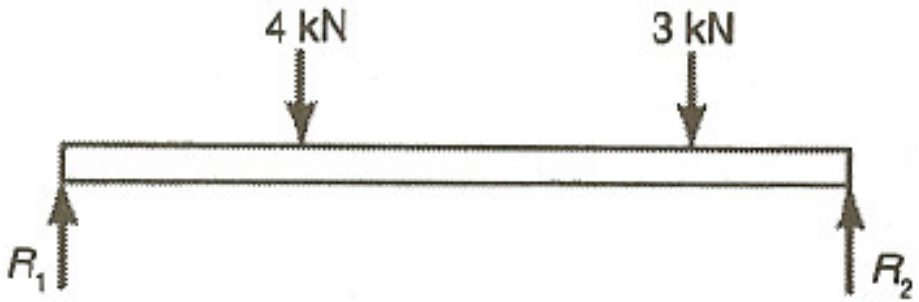
Example 3

Draw the shear and moment diagram shown in Figure E5-3.



Solution

Draw the free-body diagram of the beam.

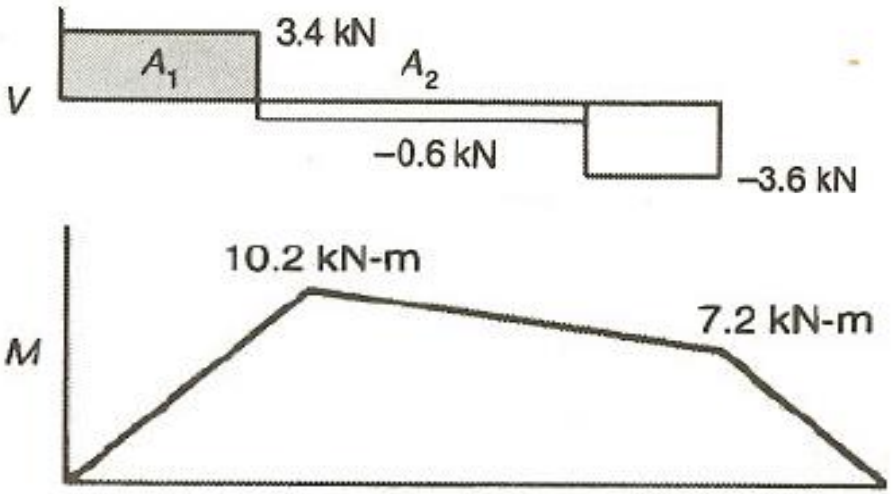


Summation of the moments about the right end

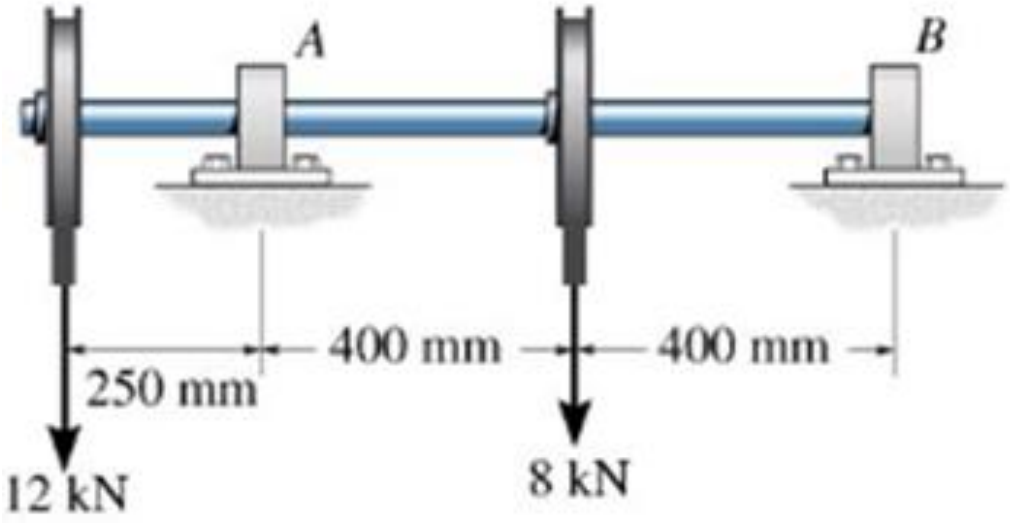
$$10R_1 = (4)(7) + (3)(2) = 34 \Rightarrow R_1 = 3.4kN$$

Summation of forces in the vertical direction

$$R_2 = 7 - R_1 = 7 - 3.4 = 3.6kN$$

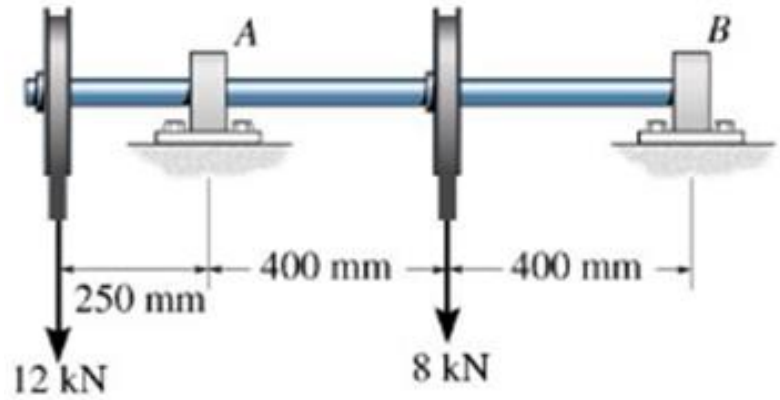


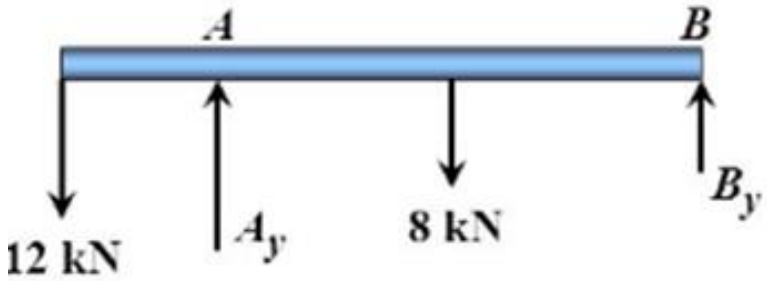
Example 4



Draw the shear force and bending moment diagrams for the beam shown above

Example 4





Free-body diagram

Reaction forces:

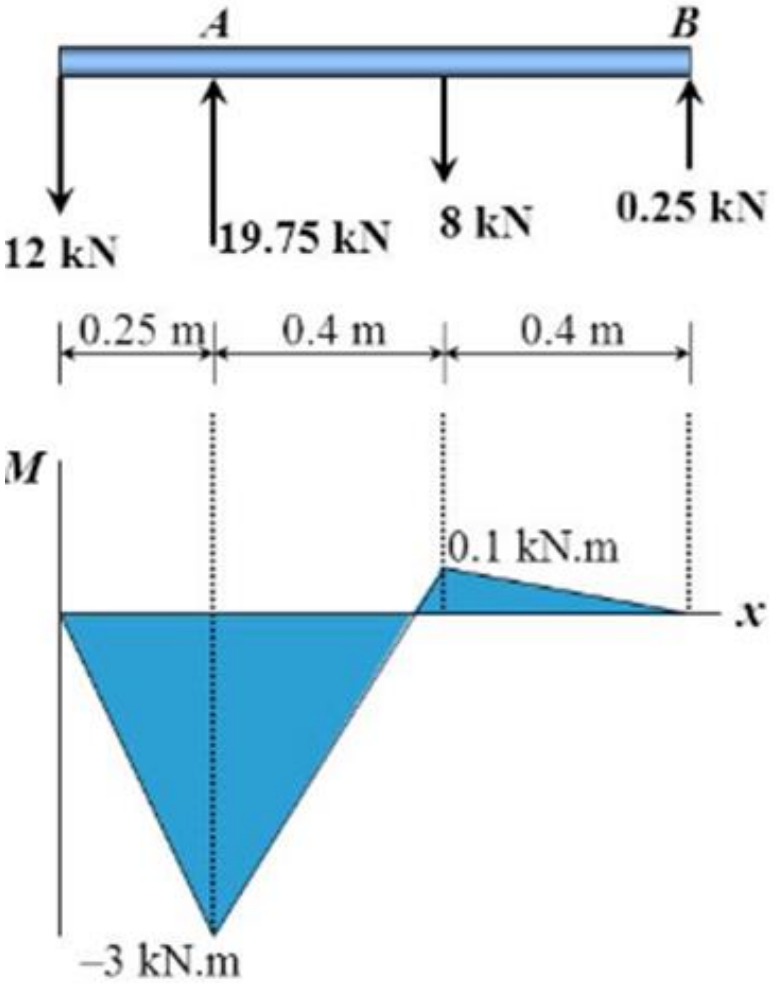
$$\Sigma M_A = 0;$$

$$B_y = \frac{(8x0.4)-(12x0.25)}{0.8} = 0.25\,\text{kN}$$

$$\Sigma F_y = 0; \quad \rightarrow \quad A_y = 12 + 8 - B_y$$

$$A_y = 19.75\ \text{kN}$$

Example 4 (Bending Moment Diagram)



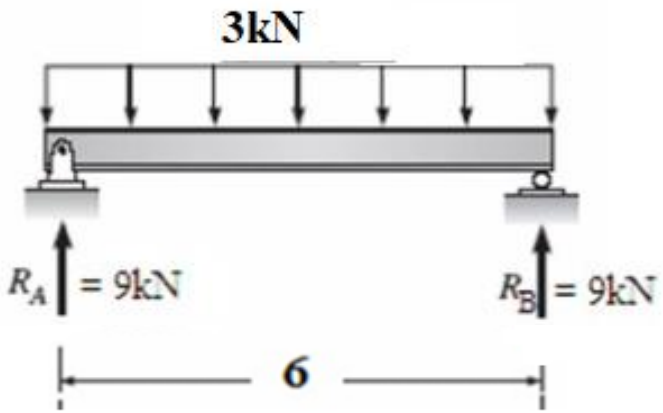
$M_0 = 0$

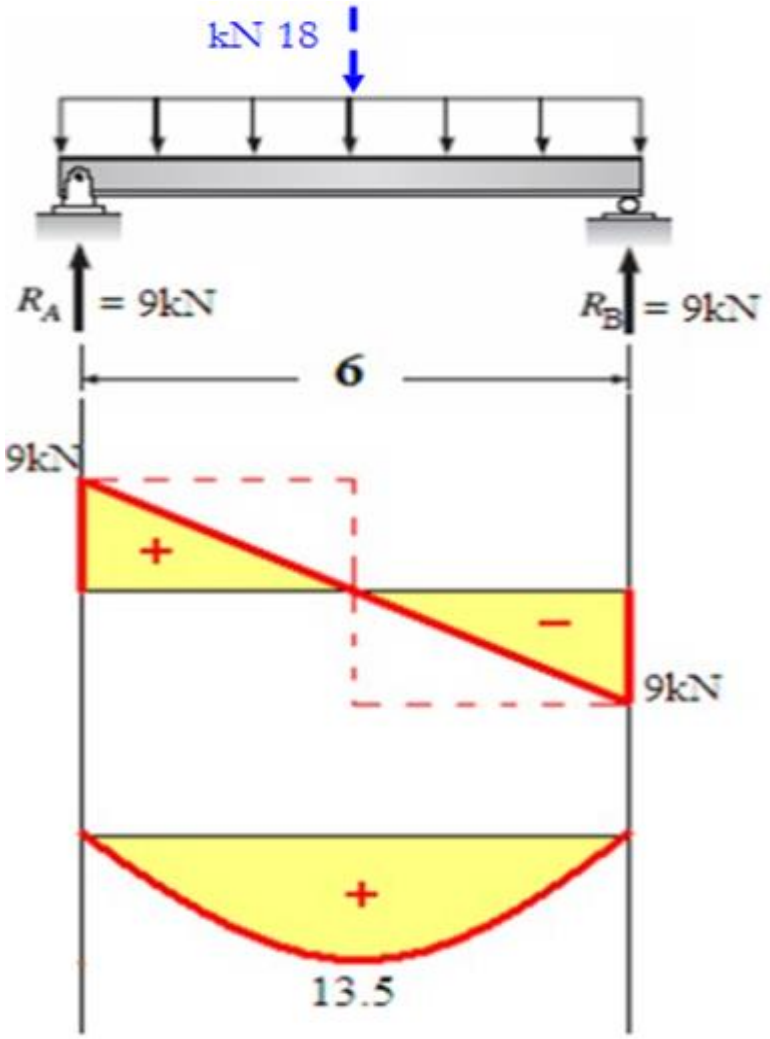
$M_{0.25} = -12 \times 0.25 = -3$ kN.m

$M_{0.65} = (-12 \times 0.65) + (19.75 \times 0.4) = 0.1$ kN.m

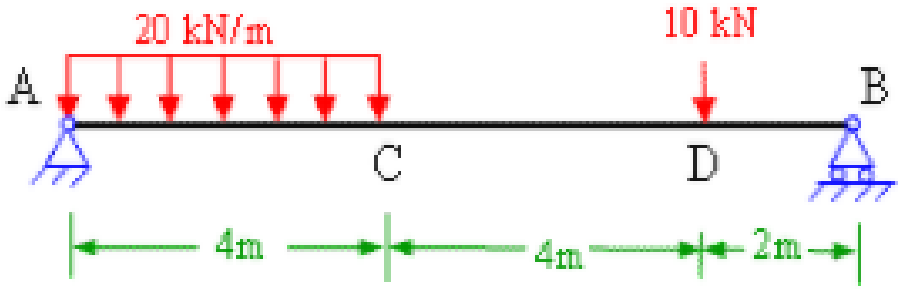
$M_{1.05} = 0$

Shear force and Bending Moment for Uniformly distributed Load





Example 5



Applying equations of static equilibrium $\sum F_A = 0;\ A_x = 0$

$$\sum F_y = 0;\ A_y + B_y - 10 - 20 \times 4 = 0$$

$$A_y + B_y = 90\ kN \qquad eq\ (1)$$

Taking moment about A

$$\sum M_A = 0;\ B_y \times 10 - 10 \times 8 - (20 \times 4) \times 2 = 0$$

$$From\ eqn\ (1)\ and\ (2), B_y = 24kN, A_y = 66kN$$

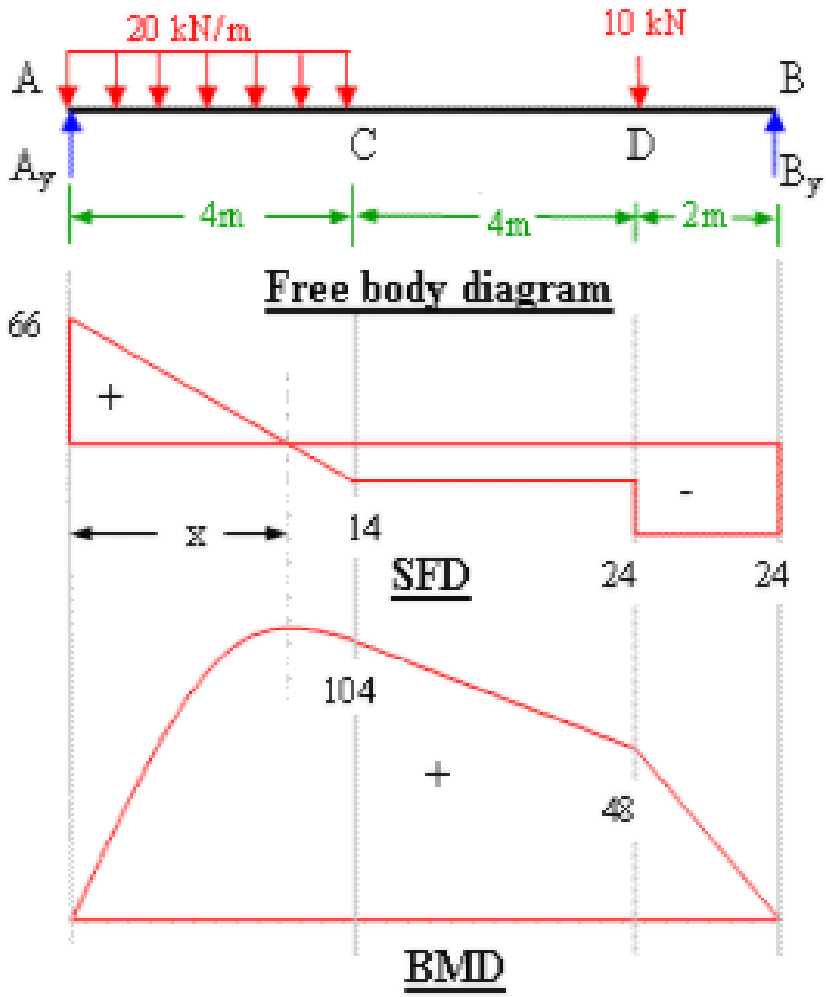
Example 5

$M_A = 0$

$M_C = 66 \times 4 - 20 \times 4 \times 2 = 104\ kNm$

$M_D = 66 \times 8 - 20 \times 4 \times 6 = 48\ kNm$

$M_B = 0$



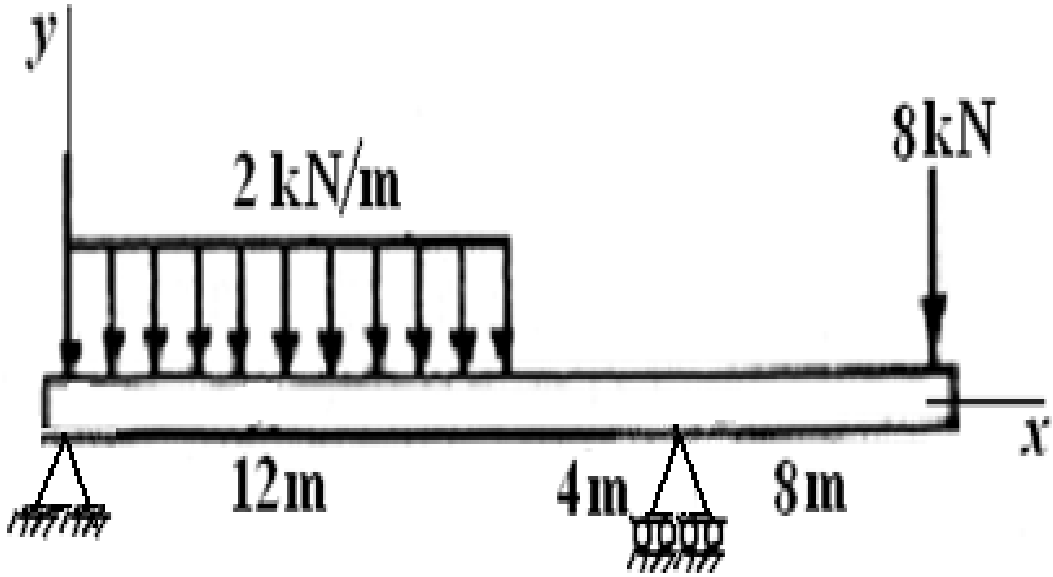
Trial Example

For the beam shown below, draw:

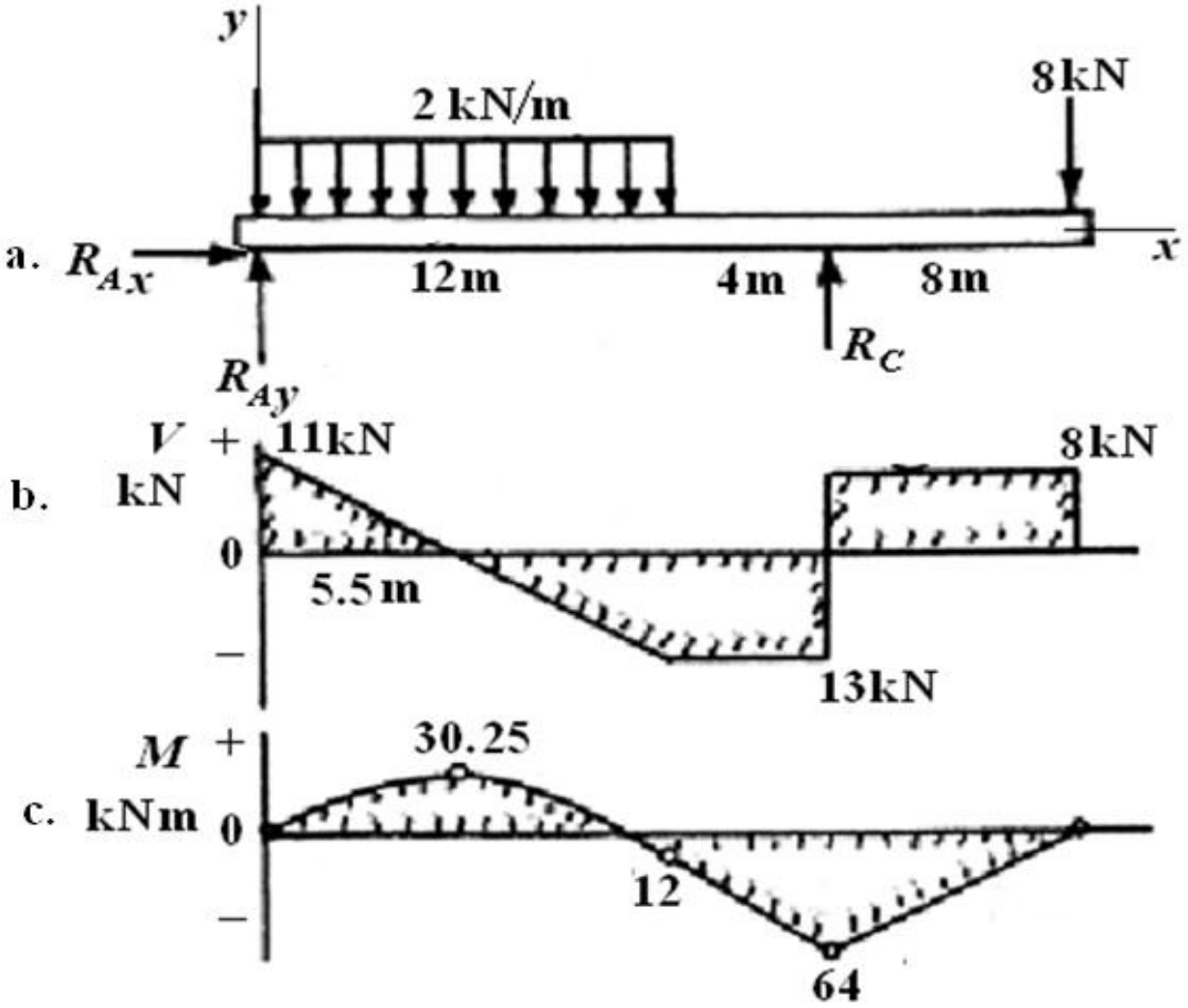
a) a free body diagram of the beam

b) the shearing force diagram of the beam

c) the bending moment diagram of the beam.



Trial Example



THANK YOU